

KUNAL HEMANT PASAD

Software Engineer | Backend Development | Distributed Systems | Scalable Applications

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EDUCATION

Rutgers, the State University of New Jersey, New Brunswick, United States

Expected: May 2027

Master of Science in Computer Science

GPA: 3.83/4.0

Relevant Coursework: Distributed Systems, Software Engineering, Database Systems, Machine Learning

Sardar Patel Institute of Technology, Mumbai, India

Dec 2021 – May 2025

Bachelor of Technology in Computer Engineering

GPA: 8.62/10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, C++

Backend: REST APIs, API Design, Microservices, FastAPI, Laravel, Node.js

Databases: PostgreSQL, MySQL, MongoDB

Technologies: Docker, Kubernetes, Caching, Asynchronous Processing

Tools: Git, CI/CD, Postman

Concepts: System Design, Backend Systems, Scalable Systems, Distributed Architectures, Data Pipelines

PROFESSIONAL EXPERIENCE

EternIQ

Jan 2025 – May 2025

Software Engineer Intern

Mumbai, India

- Engineered scalable LLM inference pipelines enabling real-time AI-driven features in backend systems
- Designed asynchronous processing workflows improving system throughput and reducing latency across inference pipelines
- Built data ingestion pipelines processing 100K+ records with validation, normalization, and schema enforcement
- Integrated REST-based inference services enabling seamless deployment of AI functionalities into backend systems

Mukesoftware IT Consultants

Jul 2023 – Feb 2024

Full Stack Web Developer Intern

Mumbai, India

- Developed backend APIs using Laravel and CodeIgniter supporting scalable applications handling 50K+ records
- Optimized SQL queries and indexing strategies reducing API response latency by 30% in production systems
- Implemented efficient data retrieval and caching strategies improving performance of frequently accessed endpoints

PROJECTS

MindSync – AI Knowledge Assistant | *FastAPI, AWS, Vector Search*

Oct 2025

- Architected backend system implementing RAG pipelines enabling scalable querying over unstructured document collections
- Developed FastAPI services supporting concurrent request handling with caching strategies improving response latency by 30%
- Designed modular pipeline enabling efficient retrieval and generation across distributed backend components under varying workloads

DeliciousBytes – Scalable Backend System | *Laravel, Docker, Kubernetes, SQL*

Dec 2025

- Designed microservices-based backend architecture supporting order workflows through 10+ REST APIs across distributed services
- Containerized services using Docker and deployed on Kubernetes enabling scalable orchestration under varying traffic loads
- Optimized database queries improving API response time by 35% and enhancing backend system performance

InstantStory – LLM Backend Pipeline | *Python, PyTorch, LLM Pipelines*

Mar 2026

- Engineered end-to-end LLM pipeline processing 13K+ samples enabling structured generation with improved coherence
- Built inference pipeline supporting real-time generation with batching and asynchronous request handling
- Optimized inference performance reducing latency by 25% through efficient prompt construction and pipeline optimization

Mutual Fund Data Pipeline & Analytics System | *Python, SQL, Pandas*

Aug 2025

- Built ETL pipeline processing 5K+ financial records with data cleaning, normalization, and validation workflows
- Designed modular data pipelines enabling efficient computation across large-scale time-series financial datasets
- Improved processing efficiency by 25% through optimized pipeline design and reduced redundant data transformations